

SUPPLEMENTARY MATERIAL

BEYOND STATISTIC PRESERVATION: CAUSAL EVALUATION OF COMPOSED FEDERATED-LEARNING DEFENSES

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This supplement holds per-cell measurement detail for the composability screen. It records numbers, not arguments: every claim these cells support is stated and qualified in the main paper, and section, table and theorem numbers without an **S** prefix refer to it, as do citations, whose full entries are in its reference list. Nothing here is required to follow the main paper’s results.

S1 THE PROSPECTIVE SUITE, ARM BY ARM

(i) **FLTrust as d_1 (7 pairs): out of scope, 2/7.** The five predicted-HIGH misses are $\text{fltrust} \rightarrow \{\text{rep}, \text{nc}, \text{krum}, \text{mkrum}, \text{tm}\}$. Their C2/C3 failures are real (FLTrust does destroy Reputation’s consensus-distance signal), and are costless because d_2 was never load-bearing: the C0 mechanism of §D.1.

(ii) **FLTrust as d_2 (6 pairs): in scope, 6/6, and the near-identity of §D.1 is visible per seed rather than only in the aggregate.** $\text{Krum} \rightarrow \text{FLTrust}$ and $\text{MKrum} \rightarrow \text{FLTrust}$ are **bit-identical** to FLTrust alone on all six runs (0.0519/0.0459/0.0453 scaling, 0.0492/0.0466/0.0426 pixel), as expected for a pass-through d_1 . $\text{NC} \rightarrow \text{FLTrust}$ is *bit-identical on the pixel attack*, where no client exceeds $\tau=5$ so $c_i \equiv 1$, and differs only within seed noise on model-scaling, where clipping actually fires. $\text{Rep} \rightarrow \text{FLTrust}$ and $\text{RFA} \rightarrow \text{FLTrust}$ likewise differ only within noise. The single genuine deviation is $\text{FG} \rightarrow \text{FLTrust}$ (0.064, per-seed up to 0.084): the one d_1 that produces exact *zeros* rather than positive rescalings, changing FLTrust’s effective client set instead of only its magnitudes, which is the Lemma 2 interaction. Every d_1 in the menu is a non-negative per-client rescaling whose positive part FLTrust cancels exactly, so only the *zero pattern* is free to matter, which is where the one deviation appears.

(iii) **Neither defense is FLTrust (7 pairs): 5/7.** Per-pair values are in Table A6; both misses are diagnosed in §D.1 and neither is a mechanism error.

S2 METRIC-SWAP ABLATION: VERIFICATION AND PER-CELL DETAIL

The mechanism is verified in code before it is claimed. An outcome alone would leave open whether the mirrors are invariant at all, so we check the statistic itself rather than only the ASR (`experiments/verify.cos.invariance.py`, `results/cos.invariance.check.json`), comparing each d_2 ’s internal state on $\{u_i\}$ against $\{c_i u_i\}$ for the c_i that NormClip and RFA actually produce on live rounds. Over 18 rounds (3 seeds $\times$ 2 upstream defenses $\times$ 3 rounds), `cos.reputation`’s weight vector agrees to $\leq 2.5 \times 10^{-7}$ and its weight *ordering* never changes (0/18), and `cos.krum` selects the same client in 18/18. The Euclidean originals do not: Reputation’s weights move by up to 0.222 and its ordering changes in 12/18 rounds; Krum’s selection changes in 6/18. The size of the disturbance tracks the realized weight ratio ρ , as Proposition 10 predicts: under NormClip ($\rho \in [1.00, 4.90]$) Reputation’s ordering changes in 3/9 rounds and Krum’s selection in 0/9; under RFA ($\rho \in [2.11, 17.38]$) they change in 9/9 and 6/9. On 1000 synthetic draws with larger rescalings the separation is cleaner: ordering changes in 0/1000 trials for `cos.reputation` against 992/1000 for Reputation, and 0/1000 against 618/1000 for the two Krum’s. The mirrors are additionally checked against the shipped implementations they stand in for (selected index identical, aggregate relative error $\approx 10^{-8}$, below the $\approx 7 \times 10^{-5}$ MPS-versus-CPU nondeterminism of the same computation), so the invariance is a property of the statistic and not of a divergent reimplement.

One prediction in the suite was genuinely two-sided, and it is the round’s only prospective positive result. `cos.krum`’s C1 verdict on model-scaling turns entirely on whether the accuracy gate is applied: 0.078 passes the numeric threshold, 0.146 accuracy fails the gate. Rather than choose a reading after the fact we froze *both*, with a resolution rule fixed in advance. Primary (accuracy-gated) $\Rightarrow$ C1 fails $\Rightarrow$ HIGH; secondary (numeric C1 only) $\Rightarrow$ the pair is certified $\Rightarrow$ LOW; and if the outcome had been LOW at accuracy ≥ 0.35 , the gate would have been wrong and our two primary labels for those cells would have been recorded as misses. Both pairs

came out **HIGH** (NormClip $\rightarrow$ cos_krum 0.508, RFA $\rightarrow$ cos_krum 0.566), refuting the secondary reading and vindicating the accuracy gate against a pre-registered alternative that would have overturned it. This is a small result but a real one: the gate is usually defended by argument, and here it beat a competing specification of itself on data neither of us had seen. We flag the weaker half honestly: NormClip $\rightarrow$ cos_krum's 0.508 sits barely above threshold on per-seed values 0.084/0.676/0.766, so that cell is threshold-noise even though its label is correct, and the RFA arm at 0.566 (per-seed 1.000/0.121/0.576) is what carries the conclusion.

The pattern holds across the suite once cells without power are excluded. Δ is only observable where the standalone d_2 genuinely suppresses the attack, since a baseline already above 0.5 has no suppression left to lose and will show $\Delta \approx 0$ whatever the transform did: a ceiling effect, not preservation. Restricting to standalone ASR < 0.5 at accuracy ≥ 0.35 leaves six cells, and measured disturbance splits them perfectly at both seed counts. At the frozen $n=3$: undisturbed (0/9) $\Delta \in [-0.027, +0.001]$, disturbed ($> 0/9$) $\Delta \in [+0.630, +0.865]$, a separation of $+0.629$. At $n=5$ after the post-hoc top-up: $[+0.001, +0.190]$ versus $[+0.718, +0.878]$, a separation of $+0.527$. Neither has any overlap, but the two extra seeds move both halves toward each other. Two of the three undisturbed cells are cos_krum on the pixel backdoor, where the mirror suppresses standalone (0.300 at 0.598 accuracy) and carries that suppression through under *both* upstreams: at $n=3$ intact (0.300, $\Delta = +0.000$ under NormClip; 0.273, $\Delta = -0.027$ under RFA), at $n=5$ degraded but still under threshold (0.490, $\Delta = +0.190$; 0.447, $\Delta = +0.147$; both at 0.61 accuracy, on new per-seed values 0.864/0.686 and 0.849/0.565). So the confirming half of the split is contributed by the invariant construction behaving as C2 predicts (which is the one thing the metric-swap arm does deliver, even though C1's failure on model-scaling denied it a frozen LOW label) but only weakly at $n=5$: exact invariance of the *selection* leaves the magnitude channel (Lemma 2) free to move Δ , and two seeds are enough to see it move. That is a limitation of this post-hoc contrast and a direct motivation for arm 3 of the ladder.

Excluded cells. All four cos_* model-scaling cells are excluded as uninterpretable (hollow standalone baseline), and the six Reputation / cos_reputation / Krum pixel cells for lack of power (standalone ≥ 0.5).

S3 OUT-OF-SAMPLE PASS VALIDATION: PER-CELL DETAIL

(A) DBA attack transfer, at two adversarial fractions. The Distributed Backdoor Attack (Xie et al., 2020) uses spatially-distributed sub-triggers at four image corners, a categorically different attack family, and is not in the development menu ($f=0.4$ is required for viability at $K=5$). We ran three PASS-predicted compositions and three FAIL controls under DBA at $N=10$, $n=5$ seeds, first at $f=0.4$ and then, to address the power limitation the first wave exposed, again at $f=0.6$ ($K \cdot f=3.0$):

Pair	Pred.	DBA $f=0.4$	DBA $f=0.6$
fg $\rightarrow$ rfa	PASS	0.109 $\pm$ 0.065	0.125 $\pm$ 0.050
fg $\rightarrow$ cm	PASS	0.050 $\pm$ 0.027	0.045 $\pm$ 0.021
rep $\rightarrow$ cm	PASS	0.041 $\pm$ 0.014	0.043 $\pm$ 0.015
nc $\rightarrow$ rep	FAIL	0.297 $\pm$ 0.224	0.378 $\pm$ 0.185
rfa $\rightarrow$ rep	FAIL	0.297 $\pm$ 0.159	0.384 $\pm$ 0.217
fg $\rightarrow$ tm	FAIL	0.049 $\pm$ 0.031	0.068 $\pm$ 0.018
fedavg (no defense)	sanity	0.236 $\pm$ 0.190	0.386 $\pm$ 0.192

The FAIL side is uninformative at both fractions, and the mechanism is known rather than inferred. The sanity baseline never clears 0.5, so a FAIL control that also sits below 0.5 carries no information: at $f=0.4$ it is 0.236 $\pm$ 0.190 and bimodal (4/5 seeds < 0.22 , one > 0.6), because all four adversarial sub-triggers must be sampled in the same round, probability $(4/10)^4 = 2.6\%$ at $K=5$; at $f=0.6$ it rises only to 0.386 (Appendix L.3 shows the residual bimodality is a per-seed admissibility latent in the distributed trigger, not purely a co-participation effect). The distributions still separate at the tails, max per-seed ASR 0.185 across the PASS pairs against 0.712 across the FAIL pairs, which is what shows DBA embeds when co-sampling succeeds; and the PASS-FAIL gap is not merely DBA's own weakness, since fg $\rightarrow$ rfa reaches 0.125 where the undefended baseline reaches 0.386, a $3\times$ suppression. The symmetric verdict: PASS predictions are confirmed robustly at both fractions, and FAIL-side falsifiability requires a DBA variant that embeds reliably before those controls mean anything.

(B) Scale and architecture transfer, and (C) the independent survivor. All predictions were pre-registered at commit ca627e5 before any of these ran, $n=5$ seeds throughout. (B) tests fg $\rightarrow$ rfa and fg $\rightarrow$ cm at $N=100$, $K=10$, $f=0.05$ on ResNet18, a $10\times$ client-pool expansion with a different architecture, both pre-registered PASS under committed scaling and committed pixel. (C) tests fg $\rightarrow$ cm, selected *solely* by criterion PASS status and independently of the 189-config payoff-matrix scan that selected rep+tm, against temporal mixing of fg/cm (50/50 at random per round) at $N=10$, $f=0.2$:

Condition	Pred.	committed_scaling	committed_pixel
<i>(B) $N=100$, $K=10$, $f=0.05$, ResNet18</i>			
fg→rfa	PASS	0.063 ± 0.031	0.069 ± 0.029
fg→cm	PASS	0.152 ± 0.118	0.141 ± 0.099
rfa alone	baseline	0.410 ± 0.386	
<i>(C) $N=10$, $f=0.2$; oracle equals the scaling column in both rows</i>			
fg→cm composed	PASS	0.107 ± 0.031	0.099 ± 0.020
fg/cm mixed 50/50	—	0.327 ± 0.417	0.679 ± 0.159

(B): all four conditions PASS, at a maximum of 0.152. Two caveats travel with that. These runs reach only ≈ 0.29 clean accuracy at 50 rounds, ResNet18 being far from converged at this budget, so they establish that *suppression transfers* across scale and architecture and **not** that the composition is deployable at $N=100$ as configured; and the two FAIL-predicted pairs at the same scale (nc→rep C2-FAIL 0.279, fg→tm C3-FAIL 0.167) again land below 0.5, so FAIL-side discrimination is unavailable at this scale and fraction for the same reason as in (A). The single-defense contrast is what sharpens the positive result: RFA alone reaches 0.410 ± 0.386 on committed scaling, bimodal across seeds, against fg→rfa’s 0.069 ± 0.031 , a $6\times$ improvement at $N=100$.

(C): composition dominates mixing, on the committed-pixel policy only. We anchor there because it is the only policy under which both conditions retain a usable model (clean accuracy 0.66 composed, 0.77 mixed), giving 0.099 ± 0.020 against 0.679 ± 0.159 , a $6.9\times$ suppression. On committed scaling the mixed model is *destroyed* (0.11 clean accuracy, CIFAR-10 chance), so its ASR is not a meaningful backdoor comparison and we exclude it from the headline. Two limits on the claim: the mixed standard deviations exceed their means, which is the per-seed instability the pixel anchor avoids, and this is independence from the survivor scan rather than full data-independence, since fg→cm also appears in the development set at 0.107 (Table A4).

S4 3-WAY COMPOSITIONS: CRITERION SCALES WHERE EXHAUSTIVE SEARCH CANNOT

Table S1: 3-way composition criterion validation (pre-registered). Max ASR = max(mean_scaling, mean_pixel) across seeds.

Triple ($d_1 \rightarrow d_2 \rightarrow d_3$)	Predicted	Max ASR	Verdict
fg→rep→rfa	LOW	0.048	CORRECT
fg→rep→cm	LOW	0.087	CORRECT
rep→fg→rfa	LOW	0.045	CORRECT
rep→fg→cm	LOW	0.136	CORRECT
fg→nc→cm	LOW	0.109	CORRECT
fg→rfa→cm	LOW	0.041	CORRECT
rfa→nc→rep	HIGH	0.894	CORRECT
fedavg→nc→rep	HIGH	0.842	CORRECT
nc→rep→tm	HIGH	0.636	CORRECT
nc→fg→tm	HIGH	0.267	MISS
Overall accuracy		9/10 = 90%	

Selection-bias note. All 6 LOW triples in Table S1 use FoolsGold or reputation as d_1 : the same successful weighting defenses from the 2-way analysis. This reflects the structural fact that these are the only defenses satisfying C1 for both attacks in our menu. The criterion’s 3-way value is not discovering new d_1 choices but correctly *filtering* from the $5 \times 6 = 30$ triples starting with FG/rep: it predicts LOW for only 6–8 of these 30 (those where the $d_2 \rightarrow d_3$ chain preserves signals), correctly rejecting the majority. A random 6-of-10 draw from our measured set matches the criterion’s mean ASR (0.078) only 0.3% of the time ($p = 0.003$, resampling test; §D).

S5 DOSE-RESPONSE DESIGN: THE FULL PRE-REGISTERED PROTOCOL

Four properties make this a dial on the theory rather than an arbitrary perturbation. (i) It is exactly Proposition 10’s form $T(u_i) = c_i u_i$ with $c_i > 0$, so every rung lies inside the theory’s scope. (ii) **The dose is ρ :** $\max_j c_j / \min_j c_j = e^{2\rho}$ exactly, the same upstream weight ratio Theorem 8 and both mechanism-preserving

regimes are stated in, verified by reading ρ back off the *transformed update norms* on live rounds, where realized ρ matches nominal to three decimals at every rung and $\text{mean}(c)$ stays in $[0.99999, 1.00002]$. (iii) $\kappa=0$ gives $c \equiv 1$: the implementation short-circuits, so the identity rung is bit-identical to d_2 standalone by construction rather than by rounding. (iv) Normalizing the *mean* to 1 holds the aggregate’s scale fixed, separating the quantity under test (disturbance of d_2 ’s *statistic*, i.e. C2) from the magnitude-contraction channel of Lemma 2 that costs accuracy; without it the ladder would confound C2 with attenuation. The permutation is keyed on (seed, round) and **not** on client identity: assigning by id would place the adversaries at a fixed end of the ladder and systematically attenuate them, manufacturing a success, whereas this keying gives every arm at a given seed and round the *identical* coefficient vector, so the arms differ in d_2 and nothing else. **dose_kappa(κ) is an instrument, not a defense**, and is never evaluated or recommended as one.

Frozen grid and arms. $\kappa \in \{0, 0.5, 1.0, 2.0\}$, i.e. $\rho \in \{1.00, 2.72, 7.39, 54.60\}$: four rungs spanning a $\approx 55\times$ ratio. Arms are selected by the paper’s own power rule (§D.2), fixed before the freeze: the ladder is informative only where standalone d_2 genuinely suppresses the attack ($\text{ASR} < 0.5$ at clean accuracy ≥ 0.35), since a baseline already above threshold has no suppression left to lose and would give a flat curve by ceiling effect. Applying that rule to the measured baselines leaves exactly four (d_2 , attack) cells, and they happen to cover three invariance classes of Proposition 10 (Table A8). The $\kappa=0$ rung is run *fresh* at all five seeds for every arm rather than reused from the published baselines, so every rung of every curve comes from one code path at one seed set; the published baselines then serve as an independent per-seed cross-check of that rung (tolerance 10^{-6}) instead of as data in it.

The abscissa is measured, not nominal. What matters to C2 is not ρ but what the rescaling does to the specific statistic d_2 reads, so we measure that separately and *before* freezing any prediction, pushing live FL updates through the shipped transform and recording each arm’s decision at every κ on the *same* raw updates (`measure_dose_disturbance.py`, `results/dose_disturbance.json`; 5 seeds $\times$ 3 rounds per attack). No ASR is computed there, which is why it can legitimately precede the freeze. It establishes three things before any outcome exists (Table A8): *there is a dose*: disturbance is exactly 0 at the identity rung and rises monotonically with κ in every non-invariant arm; *the class-(a) invariance claim holds in the shipped code*: `cos_krum`’s selection is unchanged in 0 of 120 cells at ρ up to 54.6, its cosine *scores* moving by at most 1.1×10^{-4} relative, the float32 floor for a $\approx 1.1\text{M}$ -dimensional vector, which we record rather than assert on; and *class (b) is graded rather than binary*, at roughly half the class-(c) disturbance (0.487 vs. 0.867/0.933 at $\kappa=2.0$), which is what arm 4’s prediction turns on.

Frozen predictions. The *primary* test is pooled: across all 16 (arm, κ) cells, the rise in ASR over that arm’s own $\kappa=0$ rung is predicted to be positively rank-correlated with the measured disturbance rate (Spearman, one-sided, $\alpha = 0.05$). This is the dose–response claim the design exists for, and its predictor is a *measured property of the statistic*, not the identity of the defense and not its standalone effectiveness, both held fixed within every arm. The *secondary* predictions are the per-arm shapes of Table A8, tested by a one-sided Jonckheere–Terpstra trend test across the four ordered rungs at $n=5$. For arm 3, “flat” is scored as an *equivalence* claim rather than a failed rejection: confirmed if $|\text{ASR}(\kappa=2.0) - \text{ASR}(\kappa=0)| < 0.15$ and every rung stays below 0.5. That margin is fixed and justified in advance: it is under a quarter of the +0.629 within-defense gap for Krum on record at the freeze (the post-hoc top-up later put that gap at +0.716 at $n=5$, which only makes the margin more conservative; it is not revised), and $0.300 + 0.15 = 0.450 < 0.5$, so “within the margin” and “still suppressing” coincide by design for this arm. Arm 4’s class-(b) signature is the *shallower slope*, so its criterion includes its $\kappa=2.0$ rise being smaller than both class-(c) arms’.

S6 DOSE–RESPONSE CAVEATS RECORDED BEFORE THE FREEZE

Caveats recorded in advance. (i) Arms 1–2 are on model-scaling and arms 3–4 on the pixel backdoor. This is not a free choice: on model-scaling *no* invariant d_2 suppresses at usable accuracy, that is the finding of §D.2 (`cos_krum` 0.078 at 0.146 accuracy), which is also why `cos_krum/scaling` is not an arm. “Same attack” is required *within* each ladder, which is where the causal claim lives; across arms we compare curve *shapes and slopes*, never levels. (ii) Arm 4 has thin headroom: `coord_median/pixel` sits only 0.057 below threshold, so the binary above/below-0.5 reading is weak evidence either way and we report the continuous curve with intervals. (iii) The dose is uncorrelated with adversary status by construction, but it is not *adversary-adversarial*: a dose assignment that deliberately up-weights adversaries would be a strictly harder test and is not run here. (iv) A synthetic upstream transform is not a real defense, so we test ecological validity by overlaying the two *real* upstream transforms already measured on these same cells (NormClip and RFA, §D.2) as anchor points at their own measured disturbance; if they fall far off the synthetic curve at matched disturbance, that is a finding about the instrument and is reported as one. (v) Disturbance is measured on FedAvg trajectories on one shared set of raw updates per round, which is what makes it a clean abscissa, but means the rates in Table A8 are not re-measured along each arm’s own 50-round trajectory.

S7 DOSE-RESPONSE LADDER: PER-SEED DETAIL, INTERVALS, AND THE ANCHOR RESIDUALS

Every value below is recomputed per seed by `analyze_dose_response.py` from the ladder and disturbance JSON files; nothing is transcribed. Seeds are 42–46 in order; Table S2 carries the per-rung detail.

Table S2: Per-rung mean ASR with Student-t 95% intervals at $n=5$, the measured disturbance of that arm’s statistic, mean clean accuracy, and the per-seed values. [†] fails the 0.35 accuracy gate.

arm	$\kappa(\rho)$	disturb.	mean ASR [95% CI]	acc	per-seed ASR
1 krum / scaling	0 (1.00)	0.000	0.062 [0.019, 0.105]	0.565	0.115, 0.021, 0.047, 0.057, 0.069
	0.5 (2.72)	0.600	0.011 [−0.001, 0.023]	0.652	0.000, 0.010, 0.011, 0.027, 0.007
	1.0 (7.39)	0.733	0.014 [0.003, 0.024]	0.682	0.003, 0.015, 0.021, 0.021, 0.008
	2.0 (54.6)	0.867	0.054 [0.002, 0.106]	0.619	0.005, 0.039, 0.077, 0.113, 0.037
2 reputation / scaling	0 (1.00)	0.000	0.017 [0.012, 0.022]	0.777	0.014, 0.012, 0.017, 0.023, 0.019
	0.5 (2.72)	0.867	0.022 [0.014, 0.031]	0.779	0.016, 0.018, 0.024, 0.033, 0.020
	1.0 (7.39)	0.867	0.487 [0.349, 0.624]	0.765	0.439, 0.547, 0.578, 0.557, 0.313
	2.0 (54.6)	0.933	0.779 [0.673, 0.885]	0.701	0.710, 0.843, 0.730, 0.896, 0.714
3 cos_krum / pixel	0 (1.00)	0.000	0.490 [0.058, 0.922]	0.607	0.003, 0.269, 0.630, 0.864, 0.686
	0.5 (2.72)	0.000	0.450 [0.130, 0.769]	0.569	0.764, 0.236, 0.165, 0.648, 0.435
	1.0 (7.39)	0.000	0.153 [−0.127, 0.433]	0.466	0.132, 0.090, 0.000, 0.543, 0.001
	2.0 (54.6)	0.000	0.086 [−0.100, 0.271]	0.220 [†]	0.083, 0.000, 0.000, 0.345, 0.000
4 coord_median / pixel	0 (1.00)	0.000	0.443 [0.263, 0.622]	0.767	0.395, 0.287, 0.449, 0.677, 0.405
	0.5 (2.72)	0.251	0.428 [0.241, 0.615]	0.764	0.454, 0.215, 0.430, 0.638, 0.402
	1.0 (7.39)	0.378	0.327 [0.181, 0.472]	0.749	0.351, 0.139, 0.352, 0.463, 0.328
	2.0 (54.6)	0.487	0.170 [0.113, 0.228]	0.690	0.179, 0.090, 0.204, 0.192, 0.187

The primary test, and one post-hoc sensitivity we report but do not lean on. Pooled over all 16 cells, Spearman $\rho_s = +0.351$, one-sided $p = 0.091$ (the frozen test; refuted). Dropping the four $\kappa=0$ cells, which sit at (0, 0) by construction rather than by measurement, raises this to $\rho_s = +0.780$, $p = 0.001$ over 12 cells. We were not entitled to that analysis in advance and it does not change the verdict, and its sign is in any case driven by the *invariant* arm’s ASR falling at zero disturbance being paired with the reputation arm’s rising at high disturbance, not by the fragility pattern the framework claims: arms 1 and 4 contradict it. Against the *nominal* dose κ (equivalently $\log \rho$) instead of measured disturbance, the pooled correlation is $\rho_s = -0.354$. Neither predictor works pooled.

Per-arm trend tests. Jonckheere–Terpstra across the four ordered rungs $\times 5$ seeds, one-sided for increasing trend (normal approximation; a 20,000-draw permutation p-value with a fixed seed is reported alongside by the analysis script and agrees to the third decimal in every arm): arm 1 $J = 70.0$, $z = -0.34$, $p_{\uparrow} = 0.63$; arm 2 $J = 144.0$, $z = +4.64$, $p_{\uparrow} = 1.7 \times 10^{-6}$; arm 3 $J = 29.0$, $z = -3.10$, $p_{\uparrow} = 0.999$ ($p_{\downarrow} = 9.8 \times 10^{-4}$; dropping the gate-failing $\kappa=2.0$ rung leaves $z = -2.06$, $p_{\downarrow} = 0.020$); arm 4 $J = 31.0$, $z = -2.96$, $p_{\uparrow} = 0.998$ ($p_{\downarrow} = 1.5 \times 10^{-3}$). No frozen rule tests a decreasing trend, so those $p_{\downarrow}$ values are descriptive.

Clean accuracy against dose (descriptive, no frozen rule). Arm 1 $0.565 \rightarrow 0.652 \rightarrow 0.682 \rightarrow 0.619$ ($z = +1.35$); arm 2 $0.777 \rightarrow 0.779 \rightarrow 0.765 \rightarrow 0.701$ ($z = -3.30$); arm 3 $0.607 \rightarrow 0.569 \rightarrow 0.466 \rightarrow 0.220$ ($z = -3.50$); arm 4 $0.767 \rightarrow 0.764 \rightarrow 0.749 \rightarrow 0.690$ ($z = -4.11$). This is what separates a suppression effect from a contraction artifact: arm 4 loses 0.077 of accuracy while its ASR more than halves, whereas arm 3 loses 0.39.

Anchor residuals (pre-registered caveat (iv)). Disturbance rates from `cos_invariance_check.json`, ASRs recomputed from `metric_swap/` at the frozen $n=3$ and from `metric_swap_topup/` at $n=5$; the residual is against the synthetic rung nearest in measured disturbance. NormClip→Krum 0/9 = 0.000, ASR $0.062 \rightarrow 0.062$, residual +0.001; RFA→Krum 6/9 = 0.667, 0.691 $\rightarrow 0.779$, residual +**0.767**; NormClip→Reputation 3/9 = 0.333, 0.792 $\rightarrow 0.806$, residual +**0.789**; RFA→Reputation 9/9 = 1.000, 0.882 $\rightarrow 0.895$, residual +0.116; NormClip→cos_krum 0/9, 0.300 $\rightarrow 0.490$, residual +0.000; RFA→cos_krum 0/9, 0.273 $\rightarrow 0.447$, residual −0.044. Realized upstream ρ on the same rounds: Norm-Clip mean 3.61 (max 4.90), RFA mean 11.93 (max 17.38).

S8 DECOMPOSING THE EMITTED DISPLACEMENT: RE-SELECTION AGAINST RESCALING

The Δ agg. column of Table A11 mixes two things the causal claim of §E.2 needs apart. For a selector, $\text{agg}(U) = u_{s_0}$ and $\text{agg}(T(U)) = c_{s_1} u_{s_1}$, with s_0 the client selected on the raw stack and s_1 the client

selected on the transformed one, so exactly

$$c_{s_1} u_{s_1} - u_{s_0} = \underbrace{(u_{s_1} - u_{s_0})}_{\text{re-selection}} + \underbrace{(c_{s_1} - 1) u_{s_1}}_{\text{rescaling}},$$

both components reported relative to $\|u_{s_0}\|$. The re-selection component is what a magnitude-controlled dose would still move (it takes both updates *untransformed*) while the rescaling component is what such a control closes. They are **not orthogonal**: the measured angle runs from $\cos = -0.99$ to -0.43 , mean ≈ -0.65 , so their norms need not sum to the total, and the table prints the angle rather than leaving that discrepancy unexplained. What is asserted instead is the exact vector identity above: worst residual $2.66\text{e-}07$ over all 240 (arm, family, rung, seed, round) cells, float32 noise. **This decomposes the published number, not a second similar one**: every recomputed total agrees with the corresponding Δ agg. value in `results/admission_measurement.json` to $0.00\text{e}+00$, on the same 5 seeds $\times$ 3 live rounds, the same rungs and the same configuration as that measurement. **Selectors only**: a weighted averager and a coordinate-wise order statistic emit no single selected client, so “the selected update” and hence this split are undefined for them, and `coord.median` and `reputation` do not appear. `experiments/measure_displacement_decomposition.py` trains no model per rung, computes no ASR, writes a new file (`results/displacement_decomposition.json`) and leaves every frozen artifact byte-identical.

Design	Selector	κ	total	re-selection	rescaling	cos	sel. chg.
Confounded ladder	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.637	0.685	0.333	-0.65	0.600
		1	0.731	0.851	0.685	-0.66	0.733
		2	0.880	1.675	1.674	-0.72	0.867
Instrument (Mode S)	krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.663	0.646	0.308	-0.64	0.533
		1	0.847	1.075	0.666	-0.69	0.800
		2	0.892	0.986	0.973	-0.66	0.733
Confounded ladder	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.270	0.000	0.270	—	0.000
		1	0.515	0.000	0.515	—	0.000
		2	0.834	0.000	0.834	—	0.000
Instrument (Mode S)	cos_krum	0	0.000	0.000	0.000	—	0.000
		0.5	0.395	0.000	0.395	—	0.000
		1	0.754	0.000	0.754	—	0.000
		2	1.297	0.000	1.297	—	0.000

What it says, and what it does not. Mode S into Krum at $\kappa=2$: the published 0.892 is 0.986 of re-selection and 0.973 of rescaling, so the dose moves *which* benign client is aggregated and *how large* it arrives, in comparable measure: the instrument is not quietly a magnitude-only perturbation, nor quietly a decision-only one. Of the 11/15 rounds whose selection changes there, **none** lands on an adversary; that is the quantitative form of the claim that Krum’s decision changes are benign-to-benign, and the reason the support of the admitted adversarial mass changes in **none of the 60 measured rounds** while the decision channel moves by 0.733: on this arm no adversary is admitted at *any* rung, so that count is a floor, and the arms carrying adversarial mass at baseline are reputation and `coord.median` (48 of 60 rounds each; Table A11). `cos_krum` is the informative contrast: its statistic is exactly invariant to positive rescaling, so 0/15 selections change at every rung of *both* designs and its entire displacement (1.297 under the instrument, 0.834 under the confounded ladder) is rescaling. Its fall therefore cannot travel through the decision channel at all, which corroborates §E.2’s attribution of that fall to the magnitude channel from a quantity measured independently of any ASR. **This is a decomposition of a displacement, not of an ASR effect**: it says what the dose moves in the update entering training, makes no claim about suppression, and is an analysis of runs already reported rather than a new experiment.

S9 ADVERSARY-CORRELATED DISPERSION Δ_c , MEASURED

$\Delta_c := \mathbb{E}[c_i \mid i \in A] - \mathbb{E}[c_i \mid i \notin A]$ for every per-client transform in the paper, over the same 5 seeds $\times$ 3 rounds and, within a round, the *same* raw update stack for every transform, so the rows are comparable by construction (`measure_coefficient_targeting.py`, `results/coefficient_targeting.json`). **No ASR is computed and no model is trained per transform**, which is what lets this be added without rescoring anything; the figures are **post-hoc and exploratory** and no frozen prediction depends on them. Coefficients are read back from transformed norms, $c_i = \|T(u_i)\|/\|u_i\|$, so the table measures the shipped code

path rather than the builder’s description of it. As a provenance check the synthetic families’ adversarial coefficient shares reproduce `results/admission_measurement.json` to 0.0 absolute on all 13 cells, so the rows of Table S3 may be read on the same axis as the Mode-S and Mode-A figures of §E.2.

Table S3: ρ says how much a transform spreads the coefficients; Δ_c says who receives the spread. They are not ordered together: RFA and NormClip target the adversary 2–3.6 $\times$ harder than the ladder’s most dispersed rung at ρ that is 3.7–13 $\times$ *smaller*. Means over non-degenerate rounds; the synthetic rows are attack-pooled since their coefficients do not depend on the attack.

transform	attack	ρ	$\mathbb{E}[c \mid A]$	$\mathbb{E}[c \mid \neg A]$	Δ_c
NormClip	model-scaling	4.19	0.268	0.989	−0.721
NormClip	pixel	1.00	1.000	1.000	0.000
Reputation	model-scaling	8.3×10^5	0.003	1.387	−1.383
Reputation	pixel	1.68	0.915	1.019	−0.105
FoolsGold	model-scaling	1.9×10^{12}	1.497	0.834	+0.663
FoolsGold	pixel	1.8×10^{12}	1.365	0.986	+0.378
RFA	model-scaling	14.8	0.125	1.332	−1.208
RFA	pixel	1.69	0.923	1.012	−0.089
<i>Synthetic families of §E.1–§E.2, same code path:</i>					
ladder, $\kappa=0.5/1.0/2.0$	—	2.72/7.39/54.6	—	—	−0.140/−0.240/−0.336
Mode S, all rungs	—	1.00–54.6	—	—	0.000
Mode A, $\nu=-2 \dots +2$	—	7.39–7.39	—	—	−1.138 $\dots$ + 2.469

Three readings, in decreasing confidence. (1) ρ is not a proxy for Δ_c : the two are anti-ordered on the anchors, and NormClip on the pixel backdoor has $\rho = 1$ exactly because no client exceeds $\tau=5$, so $\Delta_c = 0$ there for a reason the invariance classification cannot see. (2) The two anchors that miss the ladder are the two most extremely targeted, which explains the residuals of §E.1 without appeal to a range mismatch. (3) Δ_c is nonetheless not a predictor of suppression: RFA is the most negative of the four real transforms and `rfa→krum` fails at 0.779, while Reputation is more negative still and `rep→coord_median` suppresses. That is the two-regime structure of §C again: strong negative Δ_c can mean either annihilation or mere dilution, and Δ_c ’s magnitude alone does not say which. We report it as the axis on which the ladder’s anchors were unmatched, not as the quantity that replaces the statistic.

S10 ANTICIPATED OBJECTIONS

Five objections we expect, answered against text that is already in the paper rather than by new argument. Nothing here is a new claim; each answer is a pointer plus the scope it turns on.

1. This is small-scale federated learning: five clients, one architecture, mostly one dataset.

Agreed, and every empirical magnitude is scoped accordingly, per arm, never pooled across arms, and enumerated in Table A2 with the scale and architecture evidence in Appendix M. What does *not* depend on scale is the identification argument: any criterion whose eligibility gate is defined on the same outcome it predicts conditions on a descendant of both mechanism and effectiveness (Proposition 6), and that holds at any client count, model size or sample size. The boundary results forbid an inference; they do not measure a system.

2. The interventions are oracles: no deployed system can pin an adversary’s coefficient or read adversary identity.

Correct, and that is why Mode S is an **instrument**, not a **screen**. Mode S measures level (P4) only by reading adversary identity, which no deployable defense can, so these experiments **identify the causal channel and are not a (P4) screen**. Pinning $c=1.0$ has one purpose, to cut the confounded path of Figure 1(a) by construction, exactly as an oracle-labelled control isolates a channel in an ablation. The one control that *removes* an oracle component (score-only, which scores the transformed stack but aggregates the untransformed selected update) does not change the verdict (§E.2, Appendix S8).

3. $n=5$ seeds is small. It is: $n=5$ per arm, $n=8$ for `cos_krum`, with Student- t 95% intervals throughout and no claim that a small effect is zero. Where suppression does not move we report **no evidence of a practically meaningful change under the pre-specified ± 0.15 equivalence margin**, a margin frozen before the runs (Table A2), never a claim that suppression is unchanged. Two design choices do more for precision than seeds would: the identity rungs are imported **bit-identical per seed** (largest deviation 0.00e+00), so the contrast is within-seed rather than across noise, and the central claim is a **sign reversal** (−0.272 against +0.098 on the identical cell, Table A3), which is not a null result. Finally, Proposition 6 says the cell that would validate the criterion is *empty by design*: more seeds cannot repair it (Table A7).

4. If ASR does not move, why does a change of decision matter? Because the change of decision is what a preservation criterion *measures*, and the gap between the two is the finding. An invariance argument certifies the top three levels of Definition 2 while the security content sits at level (P4), the adversarial contribution admitted. A criterion reading the Krum arm would call that composition broken (the decision flips in up to 80% of rounds) for a composition whose suppression shows no measurable change: a false alarm in the direction that costs coverage (Proposition 7, recall bound $1/5$). The converse fails too: `cos_krum`'s selection is bit-identical at every rung yet it never suppresses (0.584 at the identity) and its ASR *falls* 0.425 across the ladder, so exact invariance of the statistic neither implies suppression nor holds it fixed (§E.2, Appendix S7).

5. The theory covers only positive per-client rescalings. Yes, and we say so wherever it is used (Proposition 10, Theorem 8, §C.2). The class is not arbitrary (it is what norm clipping, contribution weighting and reputation weighting do to a client's update) but it excludes coordinate-wise filters and any transform that changes an update's direction, and the theorem's own margin closes under heterogeneity ($m=3.24 \rightarrow 1.01$ at $\alpha=0.1$), so **it certifies nothing where heterogeneity is high**. The negative does not rest on the theory: it is a *measured* counterexample, and a narrower theory makes the counterexample sharper rather than weaker, since **the theorem certifies mechanism preservation, not security**; it bounds no ASR.

REFERENCES

Chulin Xie, Keli Huang, Pin-Yu Chen, and Bo Li. DBA: Distributed backdoor attacks against federated learning. In *International Conference on Learning Representations*, 2020.